

The Synapse - Signal Transmission

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

A synapse is the contact point between a terminal button and the next cell. Vesicles in the terminal button contain neurotransmitters. When an electrical signal arrives, vesicles fuse with the presynaptic membrane and release neurotransmitters into the synaptic cleft. They diffuse to the postsynaptic membrane and bind to transmitter-gated channels. The channels open, positive ions enter and the membrane voltage changes. The signal is therefore passed on. Afterwards, enzymes break down the neurotransmitters, and their components are taken back into the terminal button.

Task 1: Label the structures of the synapse

Complete the labelling task in the "Synapse" mode. Export the correctly labelled figure and insert it below.



Insert here

Task 2: Sequence of signal transmission

Put the sequence of processes at the synapse in the correct order. Then export the solved task and insert it below.

Insert here

Task 3: Explanation of signal transmission

Describe how a signal is transmitted from the terminal button through the synaptic cleft to the next cell.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface. There is no handwriting or other markings on the paper.